



DATA SCIENCE FIELD NOTES / 06C

CROSS TRACK INTEGRATION EVIDENCE

From problem definition to reproducible preparation and initial no-show prediction

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	HealthConnect No-Show Prediction
MILESTONE	13 September 2026 Week 6 Batch D

Educational fictional-clinic project. Not a clinical decision system or validated operational tool.

AI simulated ML Engineering integration

This record documents the AI-assisted counterpart review requested by Wilson Moses on 13 September 2026. ChatGPT simulated the ML Engineering role while developing the Data Science submission. The exchanges below summarize the implemented review; they are not verbatim meeting minutes or evidence of another human intern participating.

Required integration field	Recorded evidence
Track collaborated with	ML Engineering, simulated by ChatGPT
Project dependency	Data Science candidate must be compatible with an executable scoring component
Received	Seven-field raw schema, output contract, feature parity and rejection-test requirements
Provided	Selected no-history pipeline, threshold scenario, comparisons and model assumptions
Activity completed	Raw-input scoring, model serialization, sample batch and test suite
What changed	History removed from inference requirements; validation and uncertainty warnings added
Evidence	src/healthconnect.py; candidate_pipeline.joblib; sample_input/output.csv; test_results.csv

Review and response

Counterpart request: Can the model score an appointment without information that is unavailable at booking? Data Science response: the chosen candidate removes all history-derived predictors and excludes reminder activity and waiting time. Implementation: the scoring contract requires no history columns, and extra outcome/history fields cannot change predictions.

Counterpart request: Can preprocessing and the classifier drift apart? Response: store them together and reload the artifact for integration testing. Implementation: direct and reloaded probabilities match, and single-record scoring matches batch scoring. No preprocessing is refitted at inference.

Validation evidence and interface handoff

Counterpart request: What happens when data is malformed or a new category appears?

Response: reject invalid required values, permit documented missing distance and unknown categories, and return warnings. Implementation: explicit date parsing, domain checks and 17 executable integration tests.

Output field	Meaning
appointment_id	Original input identifier in input order
no_show_probability	Uncalibrated experimental probability estimate for the binary cohort
support_flag	1 when probability is at least 0.600; illustrative scenario only
threshold	Applied scenario threshold
model_version	healthconnect-week6-v1
warnings	Experimental status plus Sunday, missing-distance or unknown-category flags

The seven raw fields are appointment_id, booking_date, appointment_date, age, distance_to_clinic_km, appointment_type and appointment_time. Dates use month/day/year. Age is restricted to the validated 18-80 prototype scope; this is not a clinical eligibility rule. Distance may be blank, but must be finite and nonnegative when provided.

Run `python integration/test_interface.py` from the package root to reproduce the checks. Run `python src/healthconnect.py --input integration/sample_input.csv --model models/candidate_pipeline.joblib --output integration/scored_example.csv` for a sample batch. Load only the trusted local joblib artifact; serialized Python objects should not be loaded from untrusted sources.

Acceptance boundary

The evidence demonstrates local Data Science-to-interface integration. It does not demonstrate collaboration with a human intern, acceptance by the programme, an external service, production reliability or benefit to clinic attendance. Programme confirmation and Week 7 testing remain necessary.